General

- Draw horizontal tables, not vertical
- Use standard form for any value that is less than 0.01 (so for anything $\times 10^{-n}$ where $n \geq 3$)
- Use appropriate s.f.
- % error between candidate's value and theoretical value
  = (difference between values) ÷ (**theoretical** value) × 100%
- % error in quantities from instruments
  - Error in <u>single</u> reading = (<u>half</u> smallest division) for analogue instruments, and = (smallest division) for digital instruments
  - % error = (absolute error) ÷ (measured quantity) × 100%
  - Absolute error = (single error) if single reading, and = 2 × (single error) if double reading
  - Double reading in, e.g.
    - Titre (final burette reading - initial burette reading)
    - Mass of powder (mass of full container - mass of empty container)
    - Temperature change (final temperature - initial temperature)
  - Single reading from, e.g.
    - Measuring cylinder
    - Volumetric pipette
- A straight line that does not pass through the origin does **not** indicate direct proportionality
- Improvements to accuracy
  - Method to increase dependent variable to reduce percentage error, e.g. use larger mass of limiting reagent
  - $CO_2(g)$ is soluble in water, so collect in a <u>gas syringe,</u> or:
    - Heat water before collecting gas (as temperature ↑ gas solubility ↓)
    - Saturate water with CO_2 before collecting CO_2
      - Water saturated with CO_2 can still dissolve *other* gases, only not CO_2
    - Collect the gas over a dilute strong acid
    - Suggest a method to reduce the rate of reaction (lower temperature or concentration)
  - In acid + metal/carbonate, reduce amount of $H_2(g)$ or $CO_2(g)$ lost by, e.g.:
    - Using a divided flask with one reactant in each compartment and tilting the flask to start the reaction after placing the bung
    - Wrapping the solid in tissue paper before adding it to the acid so it takes time for the acid to soak through the paper and for the reaction to start
  - In heating solids, heat for a longer duration until constant mass so that all thermal decomposition gases and water of crystallization are driven off
  - Replace the lid of the crucible when cooling so that no water vapor is absorbed from the atmosphere
    - Mention: This has a very little effect because very little water vapor is absorbed
  - Measure initial temperature before <u>each</u> experiment rather than only before Experiment 1
  - Plot a graph to obtain a better estimate of the temperature change over time
  - Plot a cooling curve
  - In ΔH_{neut}, take more readings near the endpoint in order to improve the position of the intersection (of the two lines, one before and one after maximum temperature)
- Heat a solid in a crucible with the lid <u>on</u> to prevent the solid <u>spitting</u> out of the crucible
- It is not possible to accurately measure temperature changes in a solid
- When volume of liquid/aqueous limiting reagent ↑ temperature change does NOT ↑ (stays the same) because although energy change ↑, volume is proportionally greater
- Weak acids require energy to dissociate
- If asked what assumptions have been made, consider
  - Impurities
  - Other possible reactions that may happen (other than the main one)

    - Whether the reaction was completed
    - If gas collected, was any dissolved or lost?
- When question asks for 'observation**s**', write more than one observation
- It is very possible for a gas to be released without visible effervescence
- Starch indicator is blue-black in the presence of I_2, and colorless in its absence
    - In high concentrations of iodine, release of iodine from the starch-iodine complex is slow
- Roles of $Na_2S_2O_3(aq)$
    - Reducing agent, e.g. $I_2(aq)$ to $2I^-(aq)$
    - Test for acids
        - Gives pale yellow ppt. (suspension) of S(s) with $H^+(aq)$
          $S_2O_3^{2-}(aq) + 2H^+(aq) \Rightarrow S(s) + SO_2(aq) + H_2O(l)$
        - CO_3^{2-} and MAZIT metals are better tests
- $KMnO_4(s)$ exists as purple-black crystals which, on heating, undergo thermal decomposition
    - Crystals pop/crackle
    - $O_2(g)$ produced
    - Potassium manganate(VI), K_2MnO_4 and manganese(VI) oxide, MnO_2 formed
    - MnO_2 appears as black solid, and may have slight green tint from K_2MnO_4
- Aqueous manganate(VII), MnO_4^- ions are purple
- Aqueous manganate(VI), MnO_4^{2-} ions are green
- Hydrogen peroxide, H_2O_2, is commonly a strong oxidizing agent, and also a reducing agent
    - $O_2(g)$ is liberated upon most reactions of H_2O_2
    - Decomposition of H_2O_2:
      $2H_2O_2 \Rightarrow 2H_2O + O_2$, with MnO_2 or catalase catalyst or light or heat
- $Fe^{2+}(aq)$ is pale green and $Fe^{3+}(aq)$ is yellow-brown
- 'ppt.' to mean 'precipitate' is allowed

Recording readings

Instrument	**Degree of accuracy**	**Explanation**
Burette	0.05 cm^3	Typical 50.00 cm^3 burette has 0.1 cm^3 graduations
Measuring cylinder	0.5 cm^3 or 1 cm^3 (depends)	Typical 50 cm^3 measuring cylinder has 1 cm^3 graduations and 250 cm^3 measuring cylinder has 2 cm^3 graduations
Digital balance	0.01 g	Typical digital balance has a 2 d.p. display
Thermometer	0.5℃	Typical mercury-in-glass thermometer has 1℃ graduations
Digital stopwatch	0.1 s	Typical digital stopwatch is correct to the nearest 0.01 s but human reaction time introduces uncertainty

Quantitative analysis

- Titration
    - Average two concordant (within 0.10 cm^3 of each other) titres for final answer
        - Average even if equal
        - Quote to 2 d.p. (round to nearest 0.01, not nearest 0.05)
    - Aim for first two accurate titres to be concordant
    - Do **not** perform a third accurate titration if first two accurate titres are concordant
    - Methyl orange is inaccurate because
        - Color change is indistinct and difficult to see
        - Transition range is wide, so color change occurs over addition of a significant volume
- Disappearing cross experiment

- Higher-capacity beakers have larger diameters (wider)
- Measured time depends on depth of solution
  - If deeper: light from the cross has to travel through a larger depth of liquid before reaching the air => more distorted => easier for cross to disappear => shorter measured time
  - If shallower: light from the cross has to travel through a smaller depth of liquid before reaching the air => less distorted => harder for cross to disappear => longer measured time
- If higher-capacity beaker is used with all other conditions constant, depth is smaller because diameter is larger

<table>
<tr><th>Indicator</th><th>Color in acid</th><th>Color in neutral</th><th>Color in alkali</th></tr>
<tr><td>Methyl orange</td><td>Red / Pink</td><td>Orange</td><td>Yellow</td></tr>
<tr><td>Bromophenol blue</td><td>Yellow</td><td>Green / Blue-green</td><td>Blue / Purple</td></tr>
<tr><td rowspan="5">Thymol blue</td><td colspan="3">This one's a little special</td></tr>
<tr><td colspan="2">pH < 1.2 (acid)</td><td>Red</td></tr>
<tr><td colspan="2">1.2 ≤ pH ≤ 2.8 (1st transition range)</td><td>Orange</td></tr>
<tr><td colspan="2">2.8 ≤ pH ≤ 8.0 (???)</td><td>Yellow</td></tr>
<tr><td colspan="2">8.0 ≤ pH ≤ 9.6 (2nd transition range)</td><td>Green</td></tr>
<tr><td></td><td colspan="2">pH > 9.6 (base)</td><td>Blue</td></tr>
<tr><td>Thymolphthalein</td><td>Colorless</td><td>Colorless</td><td>Blue</td></tr>
<tr><td>Phenolphthalein</td><td>Colorless</td><td>Colorless</td><td>Pink</td></tr>
</table>

Qualitative analysis

- Spend time on making as many deductions as possible
- Choose gas tests based on context, e.g.:
  - metal + acid => test for H_2
  - compound + acid => test for CO_2
- Oxidizable ions
  - SO_3^{2-} ions give pungent $SO_2(g)$ which turns damp blue litmus paper red when acid is added
  - NO_2^- ions give pungent $NO_2(g)$ which turns damp blue litmus paper red when acid is added
  - $SO_2(g)$ and $NO_2(g)$ are soluble in water, so few bubbles, but odor may be present
- SO_4^{2-} ions produce a white precipitate of lead(II) sulfate with Pb^{2+}
- Lead(II) halides are also insoluble in water (colors: white; white; bright yellow)
- CuI produces a white ppt. (e.g. with Cu^{2+} + KI, but in this case orange-brown is observed from $2I^- - 2e^- \Rightarrow I_2$)
- When something is added to something, record
  - Effervescence observed?

    - Vigor of reaction if effervescence observed
    - If one of them is solid: Soluble or insoluble?
    - Color of resulting solution
    - If not heated: Heat released?
- Describe $CO_2(g)$ test correctly
    - Gas produces white precipitate…
    - …when bubbled into limewater
- Brown gas => likely $NO_2(g)$ from thermal decomposition of nitrate

Choosing a gas test (and deductions)

- Heating ionic compound in hard-glass test tube
    - Condensation at mouth of test tube => water vapor (water of crystallization or thermal decomposition of bicarbonate, HCO_3^-)
    - Cation known to be Group I => test for oxygen (thermal decomposition of nitrates)
        - Look out for brown $NO_2(g)$, same reason
    - Depending on context, test for ammonia or carbon dioxide
        - Ammonia released => ammonium cation
        - Carbon dioxide released => likely Group II carbonate (thermal decomposition)
- Adding metal to solution => test for hydrogen (acid-metal or alcohol-metal reaction)
    - If solution changes from colorless => pale blue, then metal is Cu
        - If solution is acid, then acid is HNO_3, reaction is redox (not acid-metal)
        - If solution is aqueous ionic compound, then cation is Ag^+, reaction is displacement
            - You are too poor for ions of gold and platinum
- Adding ionic compound to solution => test for carbon dioxide (acid + carbonate)
- Adding $H_2O_2(aq)$ / $MnO_2(s)$ to anything => test for oxygen (redox reaction / decomposition of H_2O_2)
- Heating anything with NaOH(aq) => test for ammonia
    - If positive, NH_4^+ present
    - If negative and Al(s) added after, test for ammonia again
        - If positive, NO_3^- or NO_2^- present
            - Dilute HCl(aq) can be used to distinguish between the two